Socioeconomic Impact Evaluation of Terrestrial Protected Areas in Madagascar based on large national surveys

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Abstract Protected areas are the most widely used instrument for biodiversity conservation, yet their socioeconomic effects on nearby populations remain contested, particularly in low income contexts where livelihoods depend heavily on natural resources. This paper evaluates the impact of terrestrial protected areas on rural household well being in Madagascar, exploiting geolocated socio-demographic surveys covering the period of rapid protected area expansion between 2008 and 2021. To support identification, we draw on an earlier survey wave from 1997 to assess the plausibility of parallel trends prior to treatment. We combine spatially derived socio environmental indicators with genetic matching and difference in differences to construct a credible counterfactual. The study follows a pre analysis plan registered before any outcome analysis, including an ex ante power calculation indicating the ability to detect effects of approximately 7.5 wealth percentiles. We find no statistically significant average effect of protected area creation on the wealth index of households located within 10 km of protected areas established after 2008. A series of robustness checks using alternative data sources and specifications supports this result. The absence of an average effect may reflect offsetting positive and negative mechanisms, heterogeneous impacts across households or sites, or limited implementation capacity in a context of weak governance and chronic underfunding of protected areas.

1 Introduction

The reconciliation between conservation and economic development has long debated in the scientific literature (Adams et al. 2004), but the issue has gained renewed prominence over the past decade with the rapid glocal expansion of protected ares (PA). This debate is particularly salient in the context of the Kunming–Montreal Global Biodiversity Framework, under which 195 signatory states committed to protecting 30 percent of terrestrial land by 2030.

In theory, PAs can affect local livelihoods through multiple and potentially opposing channels. While they are a central instrument for biodiversity conservation (Maxwell et al. 2020), their establishment may restrict access to land and natural resources that support income generating activities such as agriculture, hunting, fishing, or forest product collection. At the same time, protected areas may generate benefits through compensation schemes, employment opportunities linked to conservation or tourism, and the provision of ecosystem services such as water regulation, erosion control, or fire prevention (Kandel et al. 2022).

Despite these ambivalent mechanisms, rigorous quantitative evidence on the socioeconomic impacts of PAs remains limited, in particular in poor and fragile governance contexts. Among the 1,043 studies reviewed across 104 countries by McKinnon et al. (McKinnon et al. 2016), only 19 used quantitative methods to assess effects on material living conditions or economic well being. More recent syntheses confirm substantial heterogeneity in estimated impacts across contexts and methodologies. Focusing on 30 quantitative evaluations of household income, Kandel et al. (Kandel, Pandit, White, and Polyakov 2022) find that protected areas are associated with modest average gains, with effects that depend strongly on local conditions. This heterogeneity underscores the need for context specific evaluations using transparent and robust empirical strategies.

Madagascar provides a particularly relevant setting for such an analysis. It ranks among the poorest countries globally under SDG 1.1, with the highest share of the population living below the international poverty line (2024, pp. 298-299). During the study period, terrestrial PA coverage expanded from 3.6 percent of national territory in 2008 to 10.8 percent in 2021, while the share of the population living within 10 km of a PA rose from 9 to 28 percent. At the same time, Madagascar exhibits low state capacity (Hanson and Sigman 2021), which constrains the implementation of conservation policies and associated social measures. Combined with high rural dependence on natural resources, these features suggest that the socioeconomic impacts of protected areas may differ from those observed in less precarious institutional contexts.

Yet national scale impact evaluations remain scarce. None of the quantitative studies reviewed by McKinnon et al. (McKinnon, Cheng, Dupre, Edmond, Garside, Glew, Holland, Levine, Masuda, Miller, Oliveira, Revenaz, Roe, Shamer, Wilkie, Wongbusarakum, and Woodhouse 2016) focus on Madagascar. The only study cited by Kandel et al. (Kandel, Pandit, White, and Polyakov 2022) that includes the country relies on cross sectional municipality level data from the 1993 census (Mammides 2019), predates the major expansion of protected areas, and does not allow for before and after comparisons.

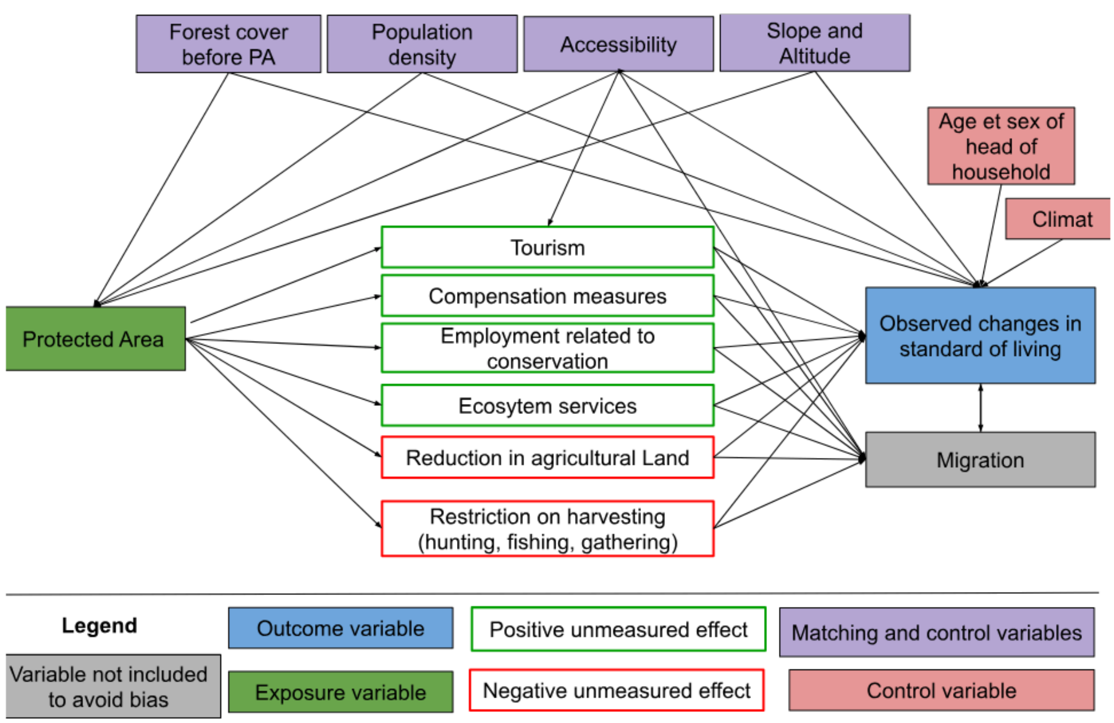
This article contributes to the literature in two ways. Empirically, it provides the first national level impact evaluation of terrestrial protected areas in Madagascar, covering 71 protected areas created between 2008 and 2021. Methodologically, it combines recent advances in matching and difference in differences methods with geolocated household survey data to construct a credible counterfactual. To enhance transparency and limit researcher discretion, the empirical strategy was fully specified in a pre analysis plan registered on the Open Science Framework prior to conducting any outcome analysis.Our analysis plan was submitted with a dated and verifiable certification on the OSF portal in Novembre 2024 and updated in March 2025.

The remainder of the paper proceeds as follows. Section Section 2, presents the conceptual framework and hypotheses. Section Section 3 describes the data sources. Section Section 5 outlines the identification strategy and econometric methods. Section Section 5 presents the econometric approaches used to assess the effect of PA on household livelihoods, the effect of PA on inequalities between households, and the heterogeneity of effects according to the type of PA governance. Section Section 6 presents our main results and proposes a series of robustness tests. Section Section 7 concludes.

2 Theory of change

Our evaluation model is based on a theory of change that links the implementation of PA(treatment) to local household well-being (the targeted results) (Figure 1) The objective here is to determine the impact of PA on observed changes in well-being. Kandel et al.(Kandel, Pandit, White, and Polyakov 2022) report a slightly positive average impact, but highlight a large heterogeneity of results across context. Several parameters are likely to influence impact, as represented graphically in Figure 1, in the form of directed acyclic graph (Hünermund and Bareinboim 2023). If the mechanisms represented affect all residents of a locality in a convergent manner, they should have a significant impact (positive or negative) on the average well being (hypothesis 1). If, on contrary, they affect them in very different ways, they may have no average impact on the well-being, but may increase inequalities (hypothesis 2)

Figure 1: Logic diagram of the theory of change tested in the study



The factors likely to lead to a decline in well-being seem particularly significant in the Malagasy context, where the population is predominantly rural and living in extreme poverty (the last assessment was in 2012, with 80.7% of the population below the $2.15 a day threshold at 2017 PPP). Six studies conducted in Madagascar between 1995 and 2006 estimated the opportunity cost of losing access to PA (slash-and-burn agriculture, hunting, gathering, timber, etc.) at between USD 39 and 177 per housejold per year (Neudert, Ganzhorn, and Waetzold 2017). Golden et al.(Golden et al. 2014) estimated that income from hunting accounted for 57% of household's cash income in areas adjacent to the Makira and Masoala PA. Another survey of people living near Makira estimated the value of pharmaceutical use at USD 30-44 per year per household, based on the subsidized price of equivalent treatments in the Malagasy market (Golden et al. 2012).

Several factors that could help improve livelihoods through conservation appear to be fragile in Madagascar, starting with tourism. Naidoo et al. (Naidoo et al. 2019) aggregate data from DHS surveys conducted between 2001 and 2011 in 34 developing countries. Their study is based on matching households near and far from PA, but with no pre-post conservation comparison. They highlight positive impacts , but only for a subset of PA 'with documented tourism'. According to their study, households living near the PA 'with tourism' are 17% wealthier and 16% less likely to be poor than similar households living far from these areas.

However, tourism in Madagascar's PA remains low. According to data from Madagascar National Parks (MNP), only 7 PA recorded more than 10,000 visitors in 2023 (with a maximum of 30,744 in Isalo), which is low compared to the average of 356,405 visitors per year and per PA recorded in 929 PA worldwide in the global study by Chung et al. (Chung, Dietz, and Liu 2018).

When new PA are created in Madagascar, compensation mechanisms for local populations remain rare, ineffective and insufficient (Rivière 2017; Bertrand et al. 2014). The most in-depth study on this subject, conducted by Poudyal et al. (Mahesh Poudyal, Jones, et al. 2018) with support from the World Bank, focuses on the Ankeniheny Zahamena Corridor (CAZ), created in 2015 to connect several existing PA. Five study sites were selected: Two adjacent to the new CAZ PA (one eligible for compensation, the other not), two adjacent to long-established PA, and one far from the forest boundary. The median cost of the conservation restriction is estimated at USD 2,375 per household per year, representing 27% to 84% of the average annual income. The amounts set aside to compensate beneficiary households were assessed to be insufficient relative to the losses incurred, and 50% of households eligible for compensation received nothing (Mahesh Poudyal, Jones, et al. 2018; Poudyal et al. 2016).

Our firts set of results therefore consists of determining whether PA in Madagascar, by limiting access to natural resources, have negative impacts on the standard of living of households living nearby, which often exceed the benefits of compensation and ecosystem services, with more adverse effects than in other countries.

The impact mechanisms represented in Figure 1 are likely to affect households differently depending on their prior characteristics, which would increase inequality (hypothesis 2). Compensation measures are generally implemented in the form of projects to promote income-generating activities (agriculture, livestock, handicrafts) in surrounding communities (Mahesh Poudyal, Rakotonarivo, et al. 2018). In the context of such development projects, individuals known as "development brokers" frequently emerge as intermediaries between local communities and implementing organizations. By mobilizing their social networks and specific skills, these brokers manage to capture a disproportionate share of the benefits of interventions, whether in form of income or access to exclusive opportunities. This dynamic can reinforce pre-existing inequalities within communities, limiting the access of the most vulnerable households to the expected benefits of compensation programs. Although tourism development is often presented as an opportunity for economic growth, it also tends to exacerbate socioeconomic inequalities, particularly in developing countries. Adeniyi et al. (Adeniyi et al. 2024) show that in Southern Africa, tourism can initially exacerbate inequalities by concentrating benefits in the most attractive regions, while leaving marginalized communities out of the economic benefits. According to Ghosh and Mitra (Ghosh and Mitra 2021), the relationship between tourism and inequality follows an inverted Kuznets curve in developing countries, when tourism remains moderate, its growth reduces inequalities, but when tourism becomes massive, further expansion worsens inequalities. Finally, Xuanming et al. (Xuanming et al. 2024) point out that while tourism helps to improve certain socioeconomic indicators, it can also generate inflationary pressures and strain local resources, particularly affecting the most vulnerable households. PA could therefore exacerbate economic inequalities among neighboring communities by creating opportunities that mainly benefit individuals with a higher educational level or a dominant position in the community, allowing them access to rents and jobs related to tourism nd associated activities.

IUCN status of PA[1] are frequently used to explain differences in effectiveness between them. For example, Naidoo et al. (Naidoo, Gerkey, Hole, Pfaff, Ellis, Golden, Herrera, Johnson, Mulligan, Ricketts, and Fisher 2019) show that multiple-use PA (statuses V and VI) tend to have more beneficial effects than strict areas (statuses I to IV), partly due to greater flexibility in integrating local needs. Beyond status alone, governance plays a central role. Eklund et al (Eklund and Cabeza 2017) highlight the importance of transparent and inclusive structures to maximize the positive effects of PA on conservation and social justice. They call for management approaches to be adapted to local contexts, with greater involvement of communities in decision-making processes, to better reconcile conservation and development objectives.

This diversity is particularly evident in Madagascar. Although governed by similar formal statuses, PA follow different paths depending on the local context and the way in which they are implemented. Froger and Méral (Froger et al. 2009) show that the early initiatives of shared governance, gradually introduced with in-depth mediation efforts, achieved encouraging results by strengthening local community support. However, from the 2000s onward, the accelerated deployment of management transfers, driven by quantitative targets, often led to hasty and less contextually adapted implementations, undermining the effectiveness of these mechanisms. These experiences demonstrate that, beyond the PA status, their establishment period, management approach, and level of community participation significantly influence their socioeconomic impacts. We therefore anticipate that The impacts of PA on well-being and inequalities are heterogeneous, and some PA with good levels of local community participation manage to generate greater benefits and distribute them more equitably (hypothesis 3).

[1]https://www.google.com/url?q=https://portals.iucn.org/library/efiles/documents/PAPS-016-Fr.pdf&sa=D&source=docs&ust=1768418520379026&usg=AOvVaw26wfaFnma2KaAr7cxRAz-5

Based on this theory of change, we define two main outcome variables to explain changes in living outcome variables to explain changes in living standards: household living standards (main variable) and the standardized Z-score of the wealth index (secondary variable).

Household living standard will be the outcome variable used to determine the overall impact of PA, and the standardized Z-score of the wealth index will explain inequalities in living standards across the localities surveyed. We also use vraibles that may be predictive of the outcome under study. The appropriate covariates for our model are variables that are likely to influence both the probability of treatment (whether a PA has been created near the household) and the outcome (household living standard and inequalities between households). The literature shows that PA tend to be created in less dense, less accessible, higher and steeper regions (Joppa and Pfaff 2010). These variables may also affect living standards: areas that are more dense, flat, low-lying and accessible (in terms of travel time and geography) tend to be wealthier (Gallup, Sachs, and Mellinger 1999). We propose five variables (forest cover in 2000, slope, elevation, population density in 2000, and accessibility in 2000).

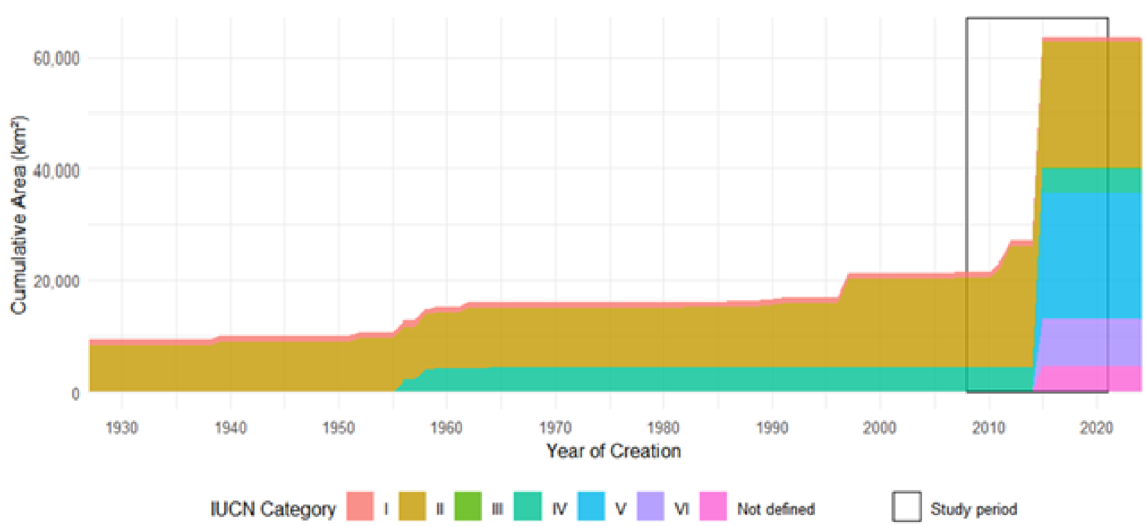
3 Data

Considering the long-term, large-scale, complex, and politically sensitive nature of the intervention to be evaluated, we use secondary data on the socioeconomic conditions of households, their geographical environment and their location in relation to PA.

3.1 Protected areas

This study evaluates the impact of terrestrial PA creation on rural household well-being between 2008 and 2021. These time frames was chosen on the basis of the availability of geolocalised data on household living conditions and coincide with a period of strong expansion of PA in the country, as shown in Figure 2

Figure 2: Evolution of protected areas in Madagascar and study period



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Notes: This graph shows the evolution of PA creation in Madagascar since its
creation in 1927 (under the colonial administration) until 2024. From 1927 until
the early 2000s, PA were characterized by strict conservation (IUCN categories I,
II and IV). At the IUCN Parks Summit in Durban in 2003, the Malagasy government
committed to trippling the area PA, which led to the creation of new PA with 28
provisional creation decrees published between April 2006 and December 2007 and
a global decree bringing the number of new PA to 97 in 2008. The final decree
was not issued until 2015, which led to the creation of new PA thereafter. The
data used to produce the graph comes from Service de la Gouvernance des Aires
Protégées (SGAP), du Ministère de l'Environnement et du Développement Durable
(MEDD)
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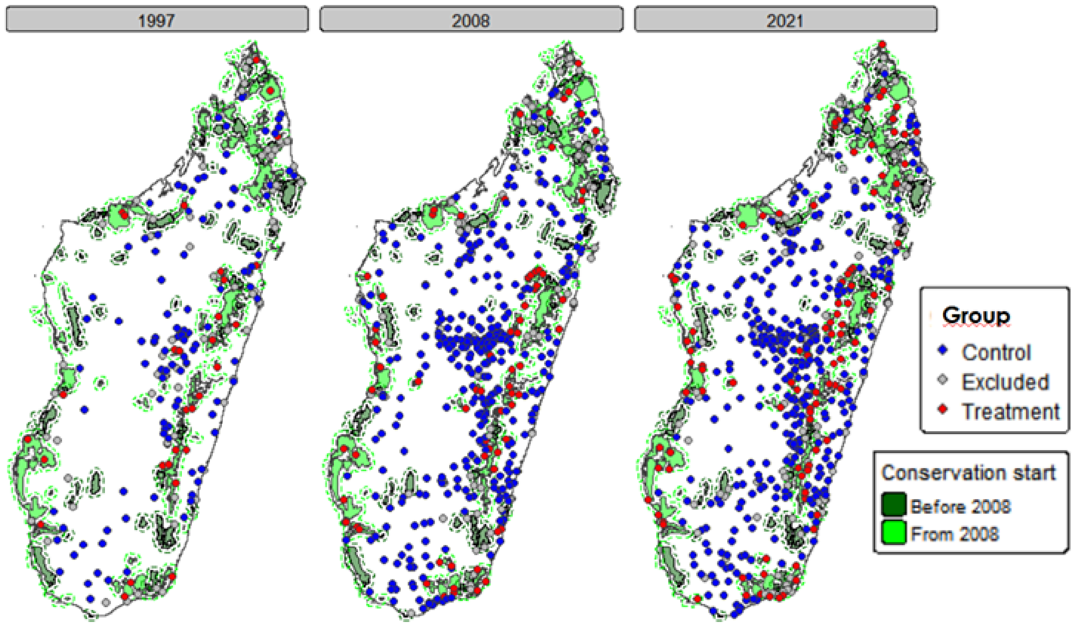
Our study analyze the impact of PA surrounding the well being over 13 years (2008-2021). We are using 2008 as the reference year. So, the population considered as treated encompasses households living in rural areas within 10 km of a PA created between 2008 and 2021, according to the GPS coordinates provided in the Demographic Health Surveys (DHS) data[2]. Households in the control group are those living in a rural area more than 10 km away from a PA created between 2008 and 2021, and they exhibit very similar characteristics or share significant traits with households in the treatment group.We decided to exclude rural populations living within 10 km of PA created before 2008, as they are considered treated before the study period; and in urban areas.

We classified PA according to their group affiliation

[2]These GPS coordinates correspond to the centroids of the enumeration areas surveyed. To protect respondent confidentiality, these coordinates are first randomly shifted using the following procedure: An offset angle between 0 and 360 is randomly drawn, and then an offset distance is randomly drawn, between 0 and 2 km in urban areas and between 0 and 5 km in rural areas. For 1% of rural clusters, the distance drawn is between 0 and 10 km (Skiles et al. 2013)

Figure 3: Classification of household clusters according their group affiliation

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Notes: Households in the treatment group are those located in clusters less than
10 km from PA created since 2008 in rural areas (shown in red) and households
in clusters more than 10 km from PA created since 2008 in rural areas (shown
in blue). Rural populations living within 10 km of PA created before 2008 and
in urban areas are excluded from the study (shown in black). The GPS data used
comes from DHS clusters.
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Table 1 show the distribution of PA (by number and area) according to the period in which they were created by final decree, taking 2008 as the reference year. In the treatment period (2008 to 2021), 137 PA were created, covering 75,191 km².

Period of creation	PA number	Area (km²)
Before 2008	34	23.939
From 2018 onwards	103	51.252

Notes: "PA number" refers to the number of protected areas created during each period.
Area is expressed in square kilometers (km²). Calculations are based on data from the World Database on Protected Areas (WDPA).

3.2 Household socioeconomic conditions

The data on household living conditions used for this study comes from surveys conducted by the "Institut National de la Statistique de Madagascar" (INSTAT) as part of the Demographic Health

Surveys (DHS)[3].This data covers a wide range of topics, including demographic characteristics, living conditions, health, education, sanitation, and hygiene. They were conducted based on surveys from 1997, 2008, and 2021, containing 650 clusters in 2021[4], 585 in 2008[5], and 268 in 1997[6].

These data are used to construct the variables for the impact assessment model. In this analysis, two outcome variables are considered: household living standards (primary variable) and inequalities in living standards at the level of the surveyed localities (secondary variable).

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• Main outcome variable: Household living standards
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The first outcome variable, household living standard, is estimated from the wealth index, calculated specifically for rural areas (variable coded hv270a in the DHS data). The wealth index is defined in the DHS data catalogue as: "A composite measure of a household's cumulative living standard. The wealth index is calculated using easy-to-collect data on a household's ownership of selected assets, such as televisions and bicycles; materials used for housing construction; and types of water access and sanitation facilities. Generated with a statistical procedure known as principal components analysis, the wealth index places individual households on a continuous scale of relative wealth. DHS separates all interviewed households into five wealth quintiles to compare the influence of wealth on various population, health and nutrition indicators. As a response to criticism that a single wealth index is too urban in its construction and not able to distinguish the poorest of the poor from other poor households, this variable provides an urban- and rural-specific wealth index" (The DHS Program/ICF 2018). As described above, we will translate this wealth index into an integer between 1 and 100, corresponding to the household's wealth percentile relative to the distribution of the whole sample.

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• Secondary outcome variable: inequality of household living standards
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In addition to the evaluation impact of PA on household living standards, we will seek to understand their influence on socioeconomic inequalities within the affected populations. To do this, we propose to use a standardized Z-Score of the wealth index, allowing for the comparison of the relative distribution of wealth around the mean within the study population, at the level of each survey cluster.

[3]The DHS surveys are based on a two-stage stratified sampling method. The population of interest is divided into 23 study areas corresponding to Madagascar's 22 regions, the capital Antananarivo(considered separately), an the Analamanga region without the capital (to isolate the impact of the capital on regional results). With the exception of the capital two strata were created in each study area. At the first level, enumeration ares (also called 'clusters') are randomly selected within each domain, with a probability proportional to the population of the cluster according to the latest census. At the second level, a sample of households is randomly selected within these clusters to participate in the survey programs

[4]657 clusters were drawn with probability proportional to size. After implementation in the field, 650 of the 657 clusters initially selected were actually visited]

[5]600 clusters were drawn with probability proportional to size. Of the 600 clusters selected, 596 could be surveyed. However, nine other clusters had invalid GPS coordinates, resulting in a total of 585 clusters for 2008

[6]270 clusters were drawn with probability proportional to size. Of the 270 clusters selected, 269 could be surveyed. However, one cluster had invalid GPS coordinates, resulting in a total of 585 clusters for 2008.

The Z-Score Z_i for each household i is calculated from the wealth index using the following formula:

$$Z_i = \frac{W_i - \mu_W}{\sigma_W}$$

(1)

where W_i is the wealth index for household i, μ_W represents the average wealth index for all households surveyed in each rural cluster, and σ_W is the standard deviation of households in the cluster.

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• Control variable
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These data also provide us with control variables: age and sex of the head of the household.These characteristics determine the economic opportunities and resource management capacity of households (Lo Bue et al. 2022). These variables reduce the unexplained variability of the model and thus enable a more accurate estimate of the treatment.

3.3 Household geophysical environment

Household geophysical environment data will be obtained using the R package mapme.biodiversity (Görgen and Bhandari 2022). This package automates the retrieval and processing of large raster-format data to produce a series of indicators applied to user-defined polygons for specified periods, which are presented and defined.

We propose the following list:

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• Forest cover rates in 2000
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correspond to the percentage coverage by vegetation with a height of 5 meters or more (Hansen et al. 2013). This variable is provided by the Global Forest Change dataset, which indicates for each pixel of one arc-second (approximately 30 meters at the equator) an estimate of forest cover in 2000 (Hansen, Potapov, Moore, Hancher, Turubanova, Tyukavina, Thau, Stehman, Goetz, Loveland, and others 2013). In Madagascar, as in other countries, PA have been preferentially created by targeting forest zones (Wilson et al. 2006; Carvalho et al. 2020). Globally, Naidoo (Naidoo, Gerkey, Hole, Pfaff, Ellis, Golden, Herrera, Johnson, Mulligan, Ricketts, and Fisher 2019) indicates that forest cover in an area is negatively correlated with the living standards of its inhabitants.

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• Slope and elevation
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are calculated using a digital terrain model from NASA-SRTM (Shuttle Radar Topography data), which provides an elevation estimate for arc-second pixels (NASA JPL 2020). Slope is measured as a percentage for each plot, while elevation is measured in meters. These topological variables influence the location of PA (Joppa and Pfaff 2010), as well as the agricultural potential of an area and therefore the living standards of rural populations (Canavire-Bacarreza and Hanauer 2013).

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• Population density in 2000
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corresponds to the estimated number of inhabitants per km² based on Worldpop data. Worldpop data provide estimates of population density for the year 2000 at a spatial resolution of about 1 km. They use modeling techniques and combines census data with various geospatial datasets(-WorldPop and CIESIN 2018).

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• Accessibility in 2000
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corresponds to the estimated travel time for households to the nearest cities, measured in minutes. These accessibility data to cities are compiled by the Joint Research Center (JRC), with 2000 as the reference year (Uchida and Nelson 2011). Accessibility to cities determines the ability to benefit from the services, products and opportunities they offer and is therefore a key factor in determining living standards in rural areas (Weiss et al. 2018; INSTAT 2020).

Each of these variables will be calculated for a circle of 10 km radius around the GPS coordinates of the survey cluster. All households in the same cluster have the same values for forest cover rate in 2000, slope and elevation, population density in 2000 and accessibility in 2000.

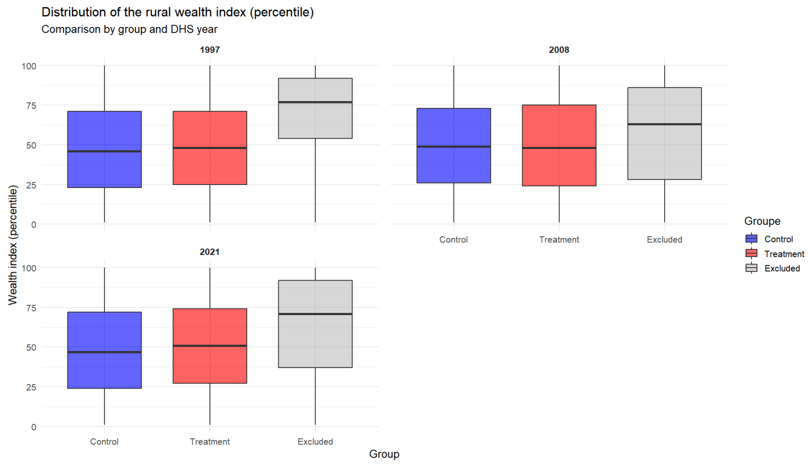
We also integrate the Standardized Precipitation Evapotranspiration Index (SPEI) (2010) for the year preceding the survey. This index is calculated based on a long-term reference (1981-2010) to quantify excess or deficit of rainfall. We use the SPEI for the year prior to the survey. The SPEI is calculated from monthly precipitation and minimum and maximum temperature data from worldclim, using an improved version of the Hargreaves method defined by (Droogers and Allen 2002), as implemented in the SPEI R package (2010). This variable will be calculated for a circle with a radius of 10 km around the GPS coordinates of the cluster.

The data from this source will constitue our matching variables. Matching covariates are used to select units not exposed to conservation (control groups) that are comparable to the exposed units (treatment group). They are appropriate for the matching process are variables likely to influence both the probability of treatment (whether a PA has been created near the household) and the outcome (household living standard and inequalities among households).

4 Descriptive statistics

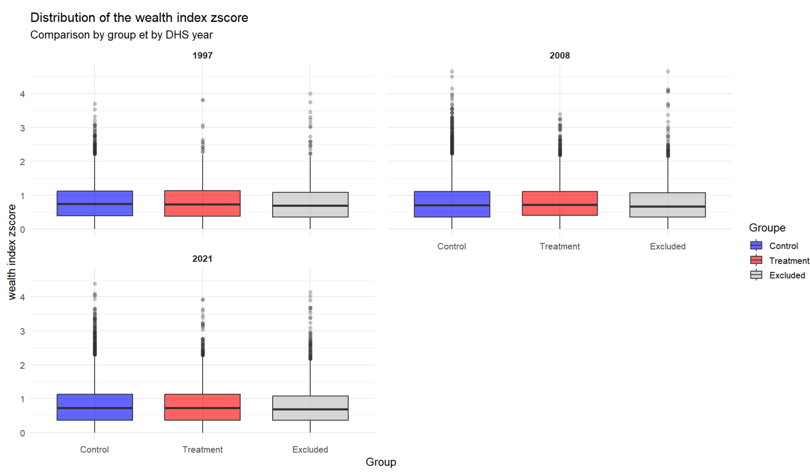
Figure 4 shows that the control and treatment groups have very similar wealth profiles, with averages between 47 and 50, and medians ranging from 46 to 51. In contrast, the excluded group is at the top of the national distribution of wealth, with an average well above 50 and a median between 63 and 77. Its distribution is generally skewed upwards (high p25). This suggests that there is no difference between the control and treatment groups; however, the standard deviations of the two groups indicate a high degree of heterogeneity in household living standards.

Figure 4: Distribution of the rural wealth index (percentile)



Notes: The graph shows boxplots of the wealth index distribution in percentiles (blue for the control group, red for the treatment group, and grey for the excluded groups). On average, for all years of the study, the wealth index of the two groups is roughly similar.

Figure 5: Distribution of the Zscore rural wealth index (percentile)



Notes: The graph shows boxplots of the zscore wealth index distribution in percentiles (blue for the control group, red for the treatment group, and grey for the excluded groups). On average, for all years of the study, the wealth index of the two groups is roughly similar.

We present the descriptive statistics for our sample in the appendix.

5 Empirical strategic

5.1 Matching methods

Our evalautive approach is based on a counterfactual measure that estimates the causal effect of the treatment, in case the implementation of PA. This approach is based on a comparison between a treatment group (household living in rural areas within or less than 10 km of PA) and a control group (households living in rural areas more than 10 km from PA created after 2008)(Schleicher et al. 2020; Desbureaux 2021). The study thus fits within the framework of Rubin's causal model (Rubin 1974), according to which there are several hypothetical outcomes depending on exposure to the treatment. To ensure comparability between groups, we apply the genetic matching (GenMatch) from the R MachtIt package to pair each unit in the treatment group with a unit in the control group that has the same observable characteristics (Ho et al. 2007). This approach uses a single distance measure, 'Mahalanobis distance matching', to quantify the similarity between the two groups of observations, while taking into account the correlations between covariates and their covariances (Diamond and Sekhon 2013). It increases the credibility of the results and reduces endogeneity issues (Ma et al. 2020). We also apply a caliper of 0.25 standard deviation of the Mahalanobis distance to avoid creating pairs with overly large differences (Rosenbaum 1983). The validity of this estimation depends on the balance between the treatment and control groups before and after matching. Before matching, we check the balance using the Standardized Mean Difference (SMD), which measures the difference between the means of the covariates in the two groups. After matching, a visual test is performed on the quality of the matching using a quantile-quantile plot.

5.2 Difference-in-difference

We assess the PA impact on household living standards and inequalities using the difference-in-difference (DID) method. The DID principle is to compare the wealth index of control and treatment households before and after the establishment of PA(note that our treatment begins in 2008).

This method relies on the parallel trends assumption,according to which treated and untreated areas evolved similarly prior to the intervention. To validate this hypothesis, we will use as a reference, among the rural households surveyed in 1997, those living in an area located within or less than 10 km of a PA created between 2008 and 2021 (placebo treatment group) and those matched to them using the method described above (placebo control group). We will conduct a placebo test by performing a DID estimation between these groups for the period 1997-2008. If the result of this placebo test is zero or statistically insignificant, the hypothesis of parallel trends is validated.

In our DID estimates, we use robust standard errors for intra-cluster correlations to manage the dependence of observations within clusters. The aim is to adjust the standard errors by taking into account the intra-cluster correlations present within groups by sampling the outcome variable (Abadie 2021). This adjustment is calculated based on the model residuals ans an estimate of the intra-cluster variance. We calculate the residuals from the regression model and then use these residuals to construct a ribust variance-covariance matrix. The matrix will adjust the intra-cluster correlation by taking into account the error structure.

The simplified equation for the difference-in-difference (Daw and Hatfield 2018) is as follows:

$$Y_{it} = \alpha_0 + \beta_1 \text{Treatment}_i + \beta_2 \text{Post}_t + \delta D_{it} + \varepsilon_{it}$$

(2)

With the parameters defined as:

Y_{it}: Wealth index for household i in year t (1997, 2008 or 2021) α_0: Intercept, representing the baseline wealth index β_1: Coefficient associated with the variable $Treatment_i$

$Treatment_i$: Binary variable that takes the value of 1 for households i affected by a PA (treatment households) and 0 for households i not affected by PA (control households)

β_2: : Coefficient associated with the variable $Post_t$

$Post_t$: Binary variable that takes the value of 1 for the post-treatment year 2021 and 0 for the pre-treatment year 2008

δ: Coefficient of interest representing the effect of PA on household living standards i in year t

D_{it}: Interaction term between $Treatment_i$ and $Post_t$; It is equal to 1 for treatment households i after the treatment year t and 0 for the other households.

ε_{it}: Error term

5.3 Robustness and sensitivity tests

We implement a series of robustness tests predefined in the pre-analysis plan published prior to the results(published on OSF in March 2025).

We re-estimate all effects using distances of 5 km and 15 km to ensure the reliability and validation of our methodology. The confidence intervals may be too wide to obtain meaningful conclusions when restricting the study area to 5 km, but comparing the coefficients will allow us to assess the consistency of the results based on the radius.

We apply Benjamini-Hochber's (Benjamini and Hochberg 1995) False Discovery Rate method to test hypothesis 2 (Pa impact on inequalities between households) and hypothesis 3 (effect on the importance of the PA governance model). These tests are performed to mitigate the risk of incorrectly inferring significant effects by controlling the average proportion of false positives among the results reported as significant. Hypothesis 2 is evaluated using the Z-score outcome variable of the wealth index, and hypothesis 3 is evaluated using the IUCN status of PA.

In the analysis, the outcome variable may be correlated with unobserved factors or shocks at the household level. Fixed effects methods can correct for many of these factors, but only repeated cross-sectional data area available here. However, this difficulty can be circumvented by using a pseudo-panel approach to estimate fixed effects models at the household cohort level (Deaton 1985)

The model is as follows:

$$\overline{Y}_{ct} = \theta_c + D_{ct} + \sum_{k=2015}^{2021} \beta \overline{Dist}_{ct} \mathbb{1}(t = k) + \delta \overline{X}_{ct} + \epsilon_i$$

(3) Y_{ct}: Average value of the household i wealth index within the cohort c at the period t

θ_c: Fixed effect controlling for the unobservable variables that remain constant over time at the cohort level

D_{ct}: Time-fixed effects at cohort level

$Dist_{ct}$: Average distance between the location of households and the boundaries of PA (where $Dist_{ct} \lessdot 10km$ within the cohort c for year t

X_{ct}: Average of the control variables (rainfall, drought) at cohort level c

ϵ_i: Error terms for individuals i

However, genetic matching and Doubly Robust DID estimation on a cross-section do not control for unobserved characteristics that may simultaneously affect PA and the outcome variabel (wealth index). To assess the robustness of the results in the face of these potential biases resulting from unobserved confounding variables, we perform a sensitivity analysis using Rosenbaum's method Rosenbaum (2002)(form: "prose"), (pp.~105–170)

6 Resultats

6.1 Matching between treatment and control

The distribution analysis of covariates prior to matching shows a significant imbalance between the treatment and control groups table). After matching, for all variables in each year, the SMD improvedcoviates improved. However, the population density varaible in 2000 remains difficult to balance perfectly. This is because PA are not randomly distributed to balance perfectly. This is because PA are not randomly distributed across the landscape: those managed for biodiversity conservation are generally located in more isolated and less populated areas, while those managed for mixed uses are surrounded by denser populationS. A Chung (Chung, Dietz, and Liu 2018) point out: "Population density around protected areas managed for mixed puposes is also higher than around protected areas managed primarily for biosiversity conservation. 'This heterogeneity between the human contexts surrounding different types of protected areas creates a lack of overlap in the distribution of population density between the tratment and control groups, making this variable difficult to balance perfectly, even after matching. Nevertheless, we achiev an acceptable balance of covariates (Annexe). The biophysical and domestic characteristic of each household in the treatment and control groups are now comparable. We apply the DID method to our study.

6.2 Overall impact on livelihoods

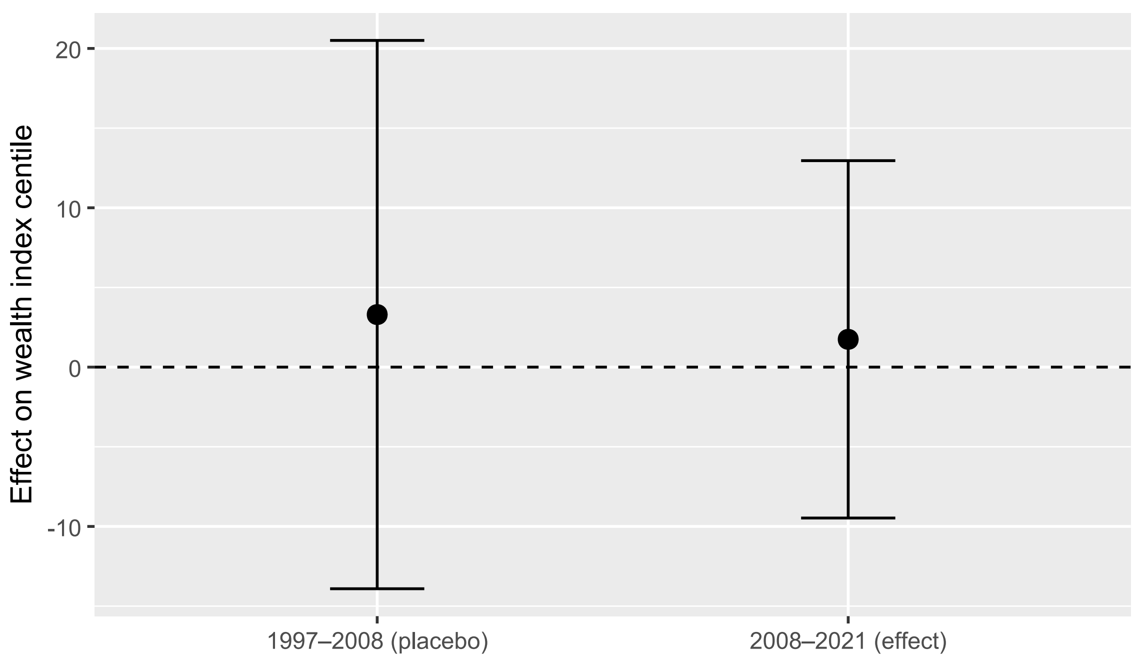
In our estimation DID, we used robust standard errors for intra-cluster correlations that correct for the dependence of observations within clusters. The results reveal that there is no statistically significant difference at the 5% level between treated and control households. Figure presents the results of the double difference estimation of the treatment groups (households within 10 km) and the control groups (households more than 10 km away) for the period 2008-2021. The principle is to compare the wealth index of control and treatment households in a period prior to (1997-2008) and after (2008-2021) the implementation of the PA.

The placebo test or pre-treatment estimate (1997-2008 period) gives an estimate with a low coefficient (+3.9 percentile) that is not significant, concluding that no difference was detected before the intervention (table). The hypothesis of parallell trends is validated, as the wealth indices of the two groups vary simularly and simultaneously before tratment (treatment in 2008)

Climatic conditions (SPEI) have no significant effect on rural wealth, and the effect of household characteristics appears to be minimal. On average, households headed by women are slightly poorer, while those with older members are richer.

Although the treatment estimate is larger in magnitude (-2.5 percentiles), the direct effect of the treatment remains insignificant. Households living near PA appear to have slightly lower level of well-being than control households. The analysis does not confirm hypothesis 1, as although the relationship is negative, the confidence interval still largely covers 0. This means that we lack sufficient precision to detect an effect, and therefore cannot reject the null hypothesis. The effect is therefore either zero or below our detection capacity. The statistical power calculation showed that the effect would need to be at least 7.8 percentiles to be detectable (Annexe)

Figure 6: Impact on well-being: 2x2 DiD impact on local population (< 10km)




```
Notes: The vertical axis shows the estimated effect of the DID on the wealth
percentile (in points). The horizontal axis shows the non significant effet.
The placebo period indicates that the error bar crosses the horizontal line at
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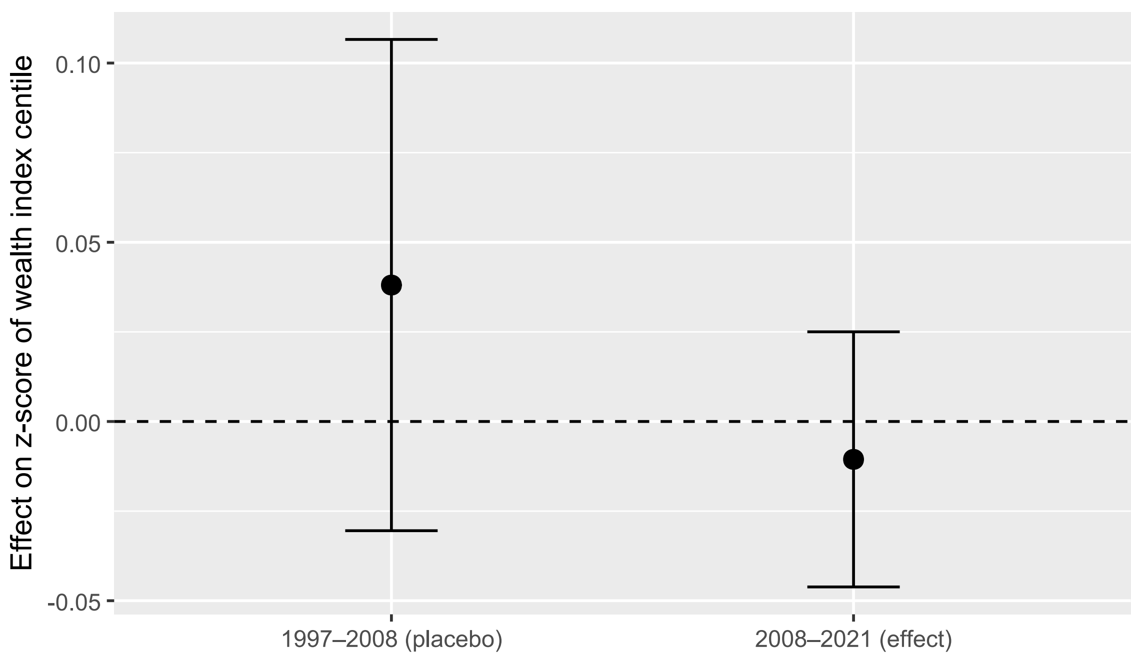
```
0(non-significant effect). During the treatment period, the clustered confidence
interval is not as wide as that of the placebo, which still encompasses 0 (non
significant effect)
```

6.3 Effect on inequalities

DID estimation shows that the average effect of being in a treated cluster did not reduce or accentuate wealth inequalities overall. The placbeo test provides an estimate with non-significant coefficient of + 0.1276 percentiles, which validates the hypothesis of parallel trends. Neither the SPEI climate variable nor the gender of the household head has a significant effect on household living standards inequality. However, only the gender of the head of household has a positive effect. During the treatment period (2008-2021), there was a general increase in inequality between households (post-coefficient = 0.7005, p<0.001), but the estimated effect remained insignificant. SPEI climate variables have a significant effect, suggesting that climte shocks disproportionately affect the most vulnerable households, accentuating inequalities. The most well-off households can cushion these shocks, creating a divergence.

Furthermore, R^2 and adjusted R^2 are close to zero and identical, indicating that the model performs poorly.

Figure 7: Impact on inequalities: 2x2 DiD impact on local population (< 10km)




```
Notes: The estimated average effect (black dot) is slightly positive for the
placebo period, but still not significant (the error bar crosses the horizontal
line). During treatment, the estimated average effect is slightly below 0,
```

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pointing to a negative effect, but as the clustered confidence interval overlaps
0, the effect is not significant
```
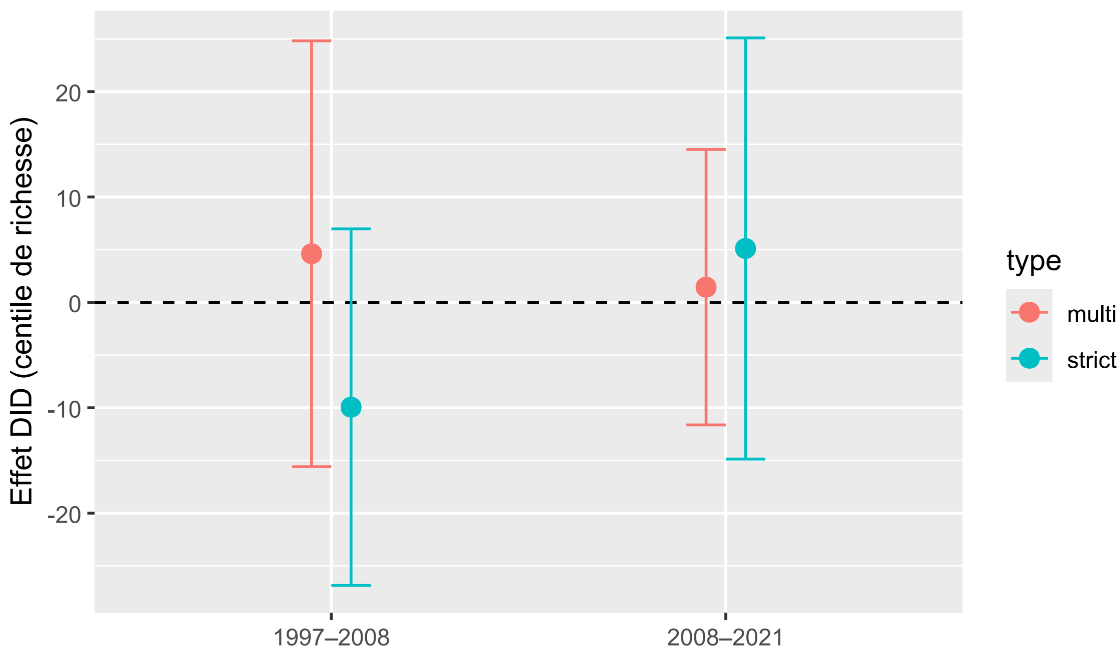

6.4 Heterogeneity

The hetereogeneous study effects aims to determine which categories of households benefit more-or less -from the creation of PA and the conditions under which this occurs. We analyzed the influence of PA governance by distinguishing between their IUCN categories 'strict' PA for those in categories I-V, and 'multi-use' PA for those in categories V-VI. The DID estimate for the period 1997-2008 shows that households living near multi-use PA experienced an increase in wealth of around five percentiles, while those living lear strict PA experienced a decrease (Figure).

Figure 8: Heterogeneity of impact on well-being




```
Notes: The graphe highlights the change in household livind standards before
and after the implementation of PA. Before PA, households living near multi-
use PA had a slightly positive wealth percentile compared to those living near
strict PA, whose living standards were rather negative. After PA, the wealth
percentile of multi-use households decreased to almost 0, while that of strict
PA increased.
```

6.5 Robustness

In this section, we discuss the sensitivity analyses and robustness checks that suppport our main results. First, to verify the robustness of our conclusions, we performed additional analyses using distances of 5 km and 15 km from the AP to define the treated group. We find that the creation of PA did not lead to significant changes in household wealth percentiles for iether distance. In neither cas is the interaction coefficient significant. Conversely, the temporal effect is consistently positive and significant, reflecting a general increase in wealth after 2008, regardless of the distance considered. Climatic variables are also statistically significant, and socio-demographic indicators consistently show that female-headed households have lower wealth and that wealth levels are positively correlated with the age of the household head. The stability of these results across the three distance thresholds suggests that climatic and socio-demographic dynamics largely dominate the effect of proximity to PA. These tests showed that our conclusions are robust to the definition of treated households.

Benjamini's (Benjamini and Hochberg 1995) False Discovery rate (FDR) method showed that, prior to multiple tetsing, the coefficients did not show a stable trend of improvement or deterioration in the wealth index. After controlling for the risk of false positives, only a marked negative effect on the wealth index remains statistically significant. This suggests a transient effect of impoverishment.

The pseudo panel approach shows that the average wealth index of household cohorts is influenced by environmental factors. The results indicate that households located at high altitudes and in densely populated areas have a higher wealth index. Forest cover has a positive and significnat effect, while slope and accessibility have a negative effect on the wealth index. In contrast, socioedemograohic variables have a weakly significant effect, suggesting that spatial and environmental conditions play a more decisive role than individual househol characteristics.

Rosenbaum's sensitivity test (Rosenbaum 1983) shows that the results are not robust in the face of unobserved assignment biases. In 1997, the absence of any systematic difference between households close to and far from PA before the introduction of PA confirms the hypothesis of parallel trends. In 2008, the test suggests a significant effect when assuming no bias ($\Gamma = 1$), but this quickly disappears as soon as a small unmeasured imbalance is introduced ($\Gamma > 1$). The Hodges-Lehmann intervals include zero and the p-value bounds exceed the 0.05 threshold. In 2021, the estimated effect is small and marginally significant, but it is not robust to hidden bias assumptions. Limiting ourselves to these three years, the observed wealth inequalities cannot be causally and robustly attributed to proximity to PA.

The DID estimates revealed that, on average, female-headed households have a lower wealth percentile (-2.307) than male-headed households (+0.7440). However, the interaction between the treatment and the gender of the household head does not reveal a signifcicant difference between men and women. IN other words, living near PA has no particular effect on gender inequality among household heads. Similarly, no interaction between the treatment and the age of the household head was evident, even though households headed by people aged 45-59 appeared to be significantly wealthier (2.920) than others.

7 Discussion and conclusion

This study aims to assess the potentially contradictory effects of creating a PA on the household living standards and socioeocnomic inequalities of rural housheolds in Madagascar. The analysis thus seeks to provide avenues for adjusting conservation policies in order to reconcile biodiversity preservation with the needs and rights of local populations.

Hypothesis 1 assumes that PA, on average, reduce the riparian household living standards. The results suggest that PA neither lead to a decline nor an increase in the wealth index of these households. However, the relationship remains neagtive, meaning the effect is either zero or below our detection capacity. The most likely explanation for this is that restrictions on the use of natural resources lead to a loss of access to supplly services and an absence of compensation mechanisms or unequal sharing of benefits. These findings highlight the need for public authorities ans conservation stkaholders to implement more inclusive measures, such as improving the sharing of tourism revenus, providing targeted cash transfers, in order to prevent increased vulnerability.

According to the literature, interventions associated with protected areas, which are supposed to improve livelihoods and economic well-being, generally tend to produce an unequal distribution of benefits. IN our case, conservation does not significantly exacerbate inequalities (hypothesis 2). THe results show that the inequalities observed are mainly due to socio-demographic and climatic dynamics. Households headed by men are wealthier than those headed by women. This reflects structural inequalities in access to resources, a lack of social recognition, and limited participation and decision-laking in the market. Conversely, households headed by older peolpe are bette off because they generally have accumulated wealth over time and stronger networks, which give them greater economic stability. Finally, climatic conditions play an important role in household living conditions: periods of drought reduce agricultural productivity and undermine livelihoods, whereas favourable conditions increase wealth. In our case, we note that climatic conditions have improved since the establisment of PA. Without inclusive policies, we would see a consolidation of 'local elites' who coul benefit more from environmental benefits. Such a situation would call for the adoption of corrective measures aimed at preventing the concentration of these benefits in the hands of a small group (for example, through local employment quotas or transparent partipatory governance), mechanisms for equitable and participatory redistribution.

Finally, hypothesis 3 concerns the heterogeneity of effects depending on the type of PA governance; The study reveals that multi-use PA (involving local communities) have positive effect on the wealth index of nearby households, whereas those living near strictly PA experience a negative effect. This difference can be explained by the fact that multi-use areas allow certain forms of sustainable resource exploitation and eocnomic activities such as ecotourism, which promotes local development. In contrast, strictly PA impose severe restrictions on access and use. Therefore, the conclusions of our study could be applied to all participatory models that have been promoted over the last fifteen years.

These findings will fuel the debate on the compatibility between conservation and poverty reduction. They will help to understand how certain institutiona; configurations or forms of governance mitigate or exacerbate the precariousness of rural households. This thus provides new insights for the empirical analysis of biodiversity protection policies in low-income countries.

8 CRediT authorship contribution statement

Iriana Razafimahenina: conceptualization - formal analysis - Methodology - writing - original draft

Florent Bédécarrats: conceptualization - Data curation - formal analysis - Funding acquisition - investigation -Methodology - Project administration - Resources - Software - Supervision - Validation - writing - original draft -

Ingrid Dallmann:conceptualization - Methodology - Supervision - Validation - Writing - review & editing

Holimalala Randriamanampisoa: conceptualization - formal analysis - Project administration - supervision - Validation- Writing - review & editing

9 Financement

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10 Declaration of interest

One of the authors is an evaluation officer at AFD, and the BETSAKA project is funded by the Evaluation department of both AFD and KfW. While the operational departments of AFD and KfW also fund conservation projects in Madagascar and other countries, the Evaluation departments operate independently. They are committed to rigorous, unbiased studies and are supervised by independent entities within both institutions.

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